Inventory Management: Part 2

Ninth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



INVENTORY MANAGEMENT

Textbook: Jay Heizer, Barry Render, Chuck Munson and Amit Sachan, Operations Management: Sustainability and Supply Chain Management, 12th Edition, Pearson India Education Services, Noida, 2017



Quantity Discount Model

- A quantity discount is a reduced price (*P*) for an item when it is purchased in larger quantities.

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (*P*)**
Initial price	1–99	Rs 1000
Discount price 1	100–1499	Rs 998
Discount price 2	1500 and above	Rs 996

- The 100 quantity and the 1,500 quantity are called *price-break quantities* because they represent the first order amount that would lead to a new lower price.

Quantity discount model

- The objective in inventory models is always to minimize total cost. Because the unit cost for the second discount in Table (last slide) is the lowest, management may be tempted to order 1,500 units.
- Placing an order for that quantity even with the greatest discount price, may not minimize total inventory cost.
- This is because holding cost increases. Thus, the major tradeoff when considering quantity discounts is between *reduced product cost* and *increased holding cost*.

Quantity discount model

- When the cost of the product is included, the total annual inventory cost can be calculated as follows:

Total annual cost = Annual setup (ordering) cost + Annual holding cost + Annual product cost

$$TC = \frac{D}{Q}S + \frac{Q}{2}IP + PD$$

where Q is the quantity ordered, D is the annual demand in units, S is setup or ordering cost per order, P is the price per unit and I is the holding cost per unit per year expressed as a fraction of price P.

Quantity discount model

- Here, the holding cost is *IP* instead of *H* as seen in the regular EOQ model. The price of the item is a factor in annual holding cost, unlike a constant holding cost.
- The price per unit changes for each quantity discount. Thus, it is common to express the holding cost as a fraction (*I*) of unit price (*P*) when evaluating costs of quantity discount schedules.
- For the quantity discount problem, EOQ formula is modified as follows:

$$Q^* = \sqrt{\frac{2DS}{IP}}$$

Quantity discount model

- Example: The annual demand is 5200 and the order cost is Rs. 2000.
- D = 5,200, S = Rs 2000, and I = 0.28.
- EOQ for Rs 996 is

$$Q^* = \sqrt{\frac{2 \times 5200 \times 2000}{0.28 \times 996}} = 274 \text{ units}$$

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (P)**
Initial price	1–99	Rs 1000
Discount price 1	100–1499	Rs 998
Discount price 2	1500 and above	Rs 996

- As 274 < 1,500 (the price-break quantity needed to receive the Rs 996 price), the EOQ for the Rs 996 price is ***not feasible*** .

Quantity discount model

- D = 5,200, S = Rs 2000, and I = 28%.
- Similarly, EOQ for Rs 998 is

$$Q^* = \sqrt{\frac{2 \times 5200 \times 2000}{0.28 \times 998}} = 273 \text{ units}$$

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (*P*)**
Initial price	1–99	Rs 1000
Discount price 1	100–1499	Rs 998
Discount price 2	1500 and above	Rs 996

- This amount ***is feasible*** because if 273 units were actually ordered, the firm would indeed receive the Rs 998 purchase price.

Quantity Discount Model

- The steps followed in quantity discount model are as follows:
  - For each discount, calculate Q^*.
  - If Q^* for a particular discount does not qualify, choose the smallest possible order size (adjusted Q^*) to avail the discount.
  - For each Q^* and adjusted value, calculate the total cost.
  - Based on the lowest total cost, make the decision.

Quantity discount model

- **Example 1:**

  Given, D = 5,200, S = Rs 2000, and I = 28%.

The quantity discount schedule is given in Table. What order quantity will minimize the total cost?

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (P)**
Initial price	1–999	Rs 1000
Discount price 1	1000–1499	Rs 960
Discount price 2	1500 and above	Rs 950

Qts ordered	Price per unit	EOQ (Q*)	EOQ adjusted (Q*)	Annual set up cost (Rs.)	Annual holding cost (Rs.)	Annual product cost (Rs.)	Total cost (Rs.)
1–999	1000	273	-----	38095.24	38220	5200000	5276315.24
1000–1499	960	279	1000	10400	134400	4992000	5136800
1500 and above	950	280	1500	6933.33	199500	4940000	5146433.33

The quantity and price that yields the lowest total cost is 1000 quantities at Rs. 960 per unit.

Quantity discount model

- **Example 2:**

  Given, D = 5,200, S = Rs 2000, and I = 28%.

The quantity discount schedule is given in Table. What order quantity will minimize the total cost?

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (P)**
Initial price	1–999	Rs 1000
Discount price 1	1000–1499	Rs 950
Discount price 2	1500 and above	Rs 940

Qts ordered	Price per unit	EOQ (Q*)	EOQ adjusted (Q*)	Annual set up cost (Rs.)	Annual holding cost (Rs.)	Annual product cost (Rs.)	Total cost (Rs.)
1–999	1000	273	-----	38095.24	38220	5200000	5276315.24
1000–1499	950	280	1000	10400	133000	4940000	5083400
1500 and above	940	282	1500	6933.33	197400	4888000	5092333.33

The quantity and price that yields the lowest total cost is 1000 quantities at Rs. 950 per unit.

Quantity discount model

- **Example 3:**

  Given, D = 5,200, S = Rs 2000, and I = 28%.

The quantity discount schedule is given in Table. What order quantity will minimize the total cost?

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (P)**
Initial price	1–999	Rs 1000
Discount price 1	1000–1499	Rs 950
Discount price 2	1500 and above	Rs 900

Qts ordered	Price per unit	EOQ (Q*)	EOQ adjusted (Q*)	Annual set up cost (Rs.)	Annual holding cost (Rs.)	Annual product cost (Rs.)	Total cost (Rs.)
1–999	1000	273	-----	38095.24	38220	5200000	5276315.24
1000–1499	950	280	1000	10400	133000	4940000	5083400
1500 and above	900	288	1500	6933.33	189000	4680000	4875933.33

The quantity and price that yields the lowest total cost is 1500 quantities at Rs. 900 per unit.

Quantity discount model

- **Example 4:**

  Given, D = 5,200, S = Rs 2000, and I = 28%.

The quantity discount schedule is given in Table. What order quantity will minimize the total cost?

Quantity Discount Schedule		
Price Range	**Quantity Ordered**	**Price per unit (P)**
Initial price	1–999	Rs 1000
Discount price 1	1000–1499	Rs 990
Discount price 2	1500–1999	Rs 970
Discount price 3	2000 and above	Rs 940

Qts ordered	Price per unit	EOQ (Q*)	EOQ adjusted (Q*)	Annual set up cost (Rs.)	Annual holding cost (Rs.)	Annual product cost (Rs.)	Total cost (Rs.)
1 to 999	1000	273	-----	38095.24	38220	5200000	5276315.24
1000–1499	990	274	1000	10400	138600	5148000	5297000
1500–1999	970	277	1500	6933.33	203700	5044000	5254633.33
2000 and above	940	282	2000	5200	263200	4888000	5156400

The quantity and price that yields the lowest total cost is 2000 quantities at Rs. 940 per unit.

Probabilistic Models and Safety Stock

- A probabilistic (statistical) model is applicable when product demand or any other variable is not known but can be specified by means of a probability distribution.
- Probabilistic models are a real-world adjustment because demand and lead time won't always be known and constant.
- An important concern of management is maintaining an adequate service level in the face of uncertain demand. The service level is the complement of the probability of a stockout.
- For instance, if the probability of a stockout is 0.05, then the service level is 0.95. Uncertain demand raises the possibility of a stockout.

Probabilistic Models and Safety Stock

- One method of reducing stockouts is to hold extra units in inventory. As mentioned earlier, such inventory is referred to as safety stock. Safety stock involves adding a number of units as a buffer to the reorder point.
- As mentioned earlier, Reorder point is given by

  ROP = Expected demand during lead time (d) + Safety stock (s_s)

- The amount of safety stock maintained depends on the cost of incurring a stockout and the cost of holding the extra inventory. Annual stockout cost is computed as

  Annual stockout costs = The sum of the units short for each demand level × The probability of that demand level × The stockout cost per unit × The number of orders per year

Probabilistic Models and Safety Stock

- ***Question:*** An optician has determined that its reorder point for eyeglass frames is 50 units. Its carrying cost per frame per year is Rs 50, and stockout (or lost sale) cost is Rs 400 per frame. The store has experienced the following probability distribution for inventory demand during the lead time (reorder period). The optimum number of orders per year is 6. How much safety stock (s_s) should the optician keep on hand?

No. of units	Probability
30	0.3
40	0.2
ROP = 50	0.3
60	0.2
70	0.1

- ***Solution:*** The objective is to find the amount of safety stock that minimizes the sum of the additional inventory holding costs and stockout costs.

Additional holding cost = Holding(carrying) cost × **Units added to ROP**

Probabilistic Models and Safety Stock

No. of units	Probability
30	0.3
40	0.2
ROP = 50	0.3
60	0.2
70	0.1

- ***Solution:*** Calculation will be done considering 0 safety stock.

For this safety stock, a shortage of 10 frames will occur if demand is 60, and a shortage of 20 frames will occur if the demand is 70. The stockout cost for zero safety stock is

Stockout cost = (10×**0**.2×400×6)+(20×**0**.1×400×6) = Rs. 9600

Safety Stock	Additional holding cost	Stockout cost	Total cost
0	0	(10×**0**.2×400×6)+(20×**0**.1×400×6) = Rs. 9600	Rs. 9600
10	50×10= Rs. 500	10×**0**.1×400×6 = Rs. 2400	Rs. 2900
20	50×20= Rs.1000	0	Rs. 1000

The safety stock with the lowest total cost is **20 frames**. Therefore, this safety stock changes the reorder point to **50 + 20 = 70 frames**.

Single-Period Model

- A single-period inventory model describes a situation in which one order is placed for a product. At the end of the sales period, any remaining product has little or no value.
  - Examples are seasonal goods, bakery goods, newspapers, and magazines.
- In other words, even though items are ordered weekly or daily, they cannot be held over and used as inventory in the next sales period.
- ***Therefore, the decision is to be made on how much to order at the beginning of the period.***

Single-Period Model

- The exact demand for such seasonal products is never known. Hence, a probability distribution related to demand is considered.
- If normal distribution is assumed, and the management stocked and sold an average (mean) of 100 Christmas trees each season, then there is a 50% chance that would stock out and a 50% chance that would have trees left over.
- To determine the optimal stocking policy for trees before the season begins, management need to know the standard deviation and consider two marginal costs: ***cost of shortage*** and ***cost of overage***

- C_s = Cost of shortage (underestimated)

  = Sales price per unit – Cost per unit
- C_o = Cost of overage (overestimated)

  = Cost per unit – Salvage value per unit (if there is any)
- The service level, the probability of not stocking out should be equal to or more than the ratio $\{C_s/(C_s + C_0)\}$

- ***Question:*** A newspaper seller sells 125 copies of a newspaper each day. He believes the sale of the newspaper is normally distributed, with a standard deviation of 15 papers. He pays Rs 3 for each paper and sells for Rs 5. He is given a Rs 1.5 credit for each unsold paper. He wants to determine how many papers he should order each day and the stockout risk for that quantity.

- ***Solution:***

C_s = Cost of shortage = Sales price per unit – Cost per unit

= Rs (5 – 3) = Rs 2

C_o = Cost of overage = Cost per unit – Salvage value per unit

= Rs (3 –1.5) = Rs 1.5

$$\frac{C_s}{C_s + C_0} = \frac{2}{2+1.5} = 0.571$$

- Now, the *Z* value has to be evaluated that yields a probability of 0.571. Using MS Excel command NORMSINV(.571), the *Z* value is 0.1789.
- The command NORMSINV(p) in MS Excel returns the value z such that, with probability p, a standard normal random variable takes on a value that is less than or equal to z.
- Hence, optimal stocking level = 125 + 0.1789×Sandard Deviation

  = 125 + 0.1789×15

  = 128 newspapers

Fixed-Period Systems

- In a fixed-period system (also called a periodic review, or *P* system), inventory is ordered at the end of a given period. On-hand inventory counted at that time.
- Only the amount necessary to bring total inventory up to a prespecified target level (*T*) is ordered.

Fixed-Period Systems

- When the time between orders (*P*) passes, an order is place to raise the inventory up to the target quantity (*T*). The amount ordered during the first period may be Q_1, the second period Q_2, and so on. The Q_1 value is the difference between current on-hand inventory and the target inventory level.
- A fixed-period system is appropriate when vendors make routine (i.e., at fixed-time interval) visits to customers to take fresh orders or when purchasers want to combine orders to save ordering and transportation costs.
- For example, a vending machine (an automatic machine that sells food or other items) company may come to refill its machines every Monday.

Fixed-Period Systems

- The disadvantage of fixed-period system is that because there is no tally of inventory during the review period. There is the possibility of a stockout during this time.
- This scenario is possible if a large order draws the inventory level down to zero right after an order is placed.
- Hence, a higher level of safety stock (as compared to a fixed-quantity system) needs to be maintained to provide protection against stockout.

Inventory Control Techniques

- Inventory control involves monitoring stock usage, movement, and storage in a business.
- It is carried out whether the inventory is located at its warehouse or in different locations.

Inventory Control Techniques

ABC Analysis

- ABC analysis divides on-hand inventory into three classifications on the basis of annual dollar volume. ABC analysis is an inventory application of Pareto principle.
- **Pareto Principle** (80-20 rule): It states that 80% of outcomes come from 20% of causes.
- The idea is to establish inventory policies that focus resources on the few critical inventory parts and not the many trivial ones.
- It is not realistic to monitor inexpensive items with the same intensity as very expensive items.

Inventory Control Techniques

ABC Analysis

- **A class:** expensive, high-class items with tight controls and small inventories. Such items may represent only about 15% of the total inventory items. They represent 70% to 80% of the total dollar usage.
- **B class:** average-priced, mid-priority items with medium sales volume and stocks. These items may represent about 30% of inventory items and 15% to 25% of the total value.
- **C class:** low-value, low-cost items with high sales and huge inventories. This represents only 5% of the annual dollar volume but about 55% of the total inventory items.

Inventory Control Techniques

- **Benefits of ABC Analysis**
  - Increased Inventory Optimization, Improved Inventory Forecasting, Better Pricing, Informed Supplier Negotiations, Better Customer Service, Strategic Resource Allocation

- **Limitations of ABC Analysis**
  - Risk of loss and theft: C class items may get lost.
  - C class item may have less price but may be crucial
  - Low information extraction: We may have insufficient data for B and C class.

Inventory Control Techniques

HML Analysis

- This method classifies inventory according to a product's unit price; ABC analysis does it on the basis of total cost usage. HML inventory method lists down products or items under the following classifications:

High Cost (H): items with high unit value.

Medium Cost (M): items with medium unit value.

Low Cost (L): items with low unit value.

Inventory Control Techniques

VED Analysis

- It is an inventory analysis method whose classification is dependent on the user's experience and perception.
- It classifies inventory according to the relative importance of certain items to other items, like spare parts.
- While performing the VED analysis, items are organized into three categories:

  Vital: inventory that need to be kept in stock consistently.

  Essential: keeping a minimum supply of this inventory is enough.

  Desirable: optional, operations can run with or without it.

Inventory Control Techniques

SDE Analysis

- The method classifies inventory based on the availability (easily available or scarce) of an item.
- It also considers the number of days of its lead time.
- It classifies inventory on three different levels:

  Scarce (S): usually imported items that require longer lead time.

  Difficult (D): items that require more than a fortnight to be available but less than six months lead time.

  Easily Available (E): readily available items.

Inventory Control Techniques

FSN Analysis

- It classifies inventory based on quantity, consumption rate, and frequency of issues and uses.
- This analysis divides the inventory based on their speed or rate of utilization, their consumption rate, and average stay.
- FSN stands for

Fast Moving (F): items that are frequently issued/used.

Slow Moving (S): items that are used less for a certain period.

Non-Moving (N): items that are not used for more than a particular duration.

Inventory Control Techniques

GOLF Analysis

- It is carried out mainly on the source of materials.
- There are many imported items which are channelized through the different trading corporations.
- Certain special procedures are required to be followed for procuring such items. As such, the ordinary procedures of inventory control may not work in respect of these items.
- Items which are available indigenously are treated differently on the basis of their sources, i.e. whether local or from very distant towns or from places where they been specially manufactured.
- GOLF stands for **Government controlled, Open market, Local supplies and Foreign market**.

Inventory Control Techniques

XYZ Analysis

- This analysis is a way to classify inventory items according to variability of their demand. Three classes of this analysis are

  X: Very little variation: X items are characterized by steady turnover over time. Future demand can be reliably forecast.

  Y: Some variation: Although demand for Y items is not steady, variability in demand can be predicted to an extent. This is because demand fluctuations are caused by known factors, such as seasonality, product lifecycles, competitor action or economic factors.

  Z: The most variation: Demand for Z items can fluctuate strongly or occur irregularly. There is no trend or predictable causal factors, making reliable demand forecasting impossible.

Inventory Control Techniques

SOS Analysis

- SOS stands for **Seasonal items** and **Off-Seasonal items**
- Seasonal items are available only for a limited period. It should be procured and stocked for meeting the needs for the entire year. The prices of the seasonal items which are available throughout the year are generally less during the harvest season.
- Agricultural items may be classified by this analysis.
- It may be advantageous to buy seasonal item at low prices and keep inventory or buy at high price during off seasons

Combination of Different Inventory Control Techniques

- ABC with XYZ classification
- ABC with VED classification
- XYZ with FSN classification
- XYZ with VED classification